# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 9 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A coin is tossed 100 times and a head appears 46 times. The experimental probability of a head is: 1

**(a)** $\frac{23}{51}$ **(b)** $\frac{23}{50}$ **(c)** $\frac{12}{25}$ **(d)** $\frac{27}{50}$

**2.** The 6th term of the sequence $1, 4, 9, 16, \ldots$ is: 1

**(a)** $35$ **(b)** $36$ **(c)** $37$ **(d)** $42$

**3.** The radius of a circle of diameter $18$ cm is (in cm): 1

**(a)** 9 **(b)** 8 **(c)** 11 **(d)** 10

**4.** The area of a rectangle of length $15$ cm and breadth $9$ cm is (cm$^2$): 1

**(a)** 136 **(b)** 135 **(c)** 137 **(d)** 134

**5.** A fair die is rolled once. The probability of getting a number greater than 4 is: 1

**(a)** $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{6}$

**6.** For the sequence $t_n = 3n + 2$, the value of $t_4$ is: 1

**(a)** $14$ **(b)** $13$ **(c)** $15$ **(d)** $16$

**7.** The perpendicular from the centre of a circle to a chord of length $16$ cm bisects it. Each half is (in cm): 1

**(a)** 9 **(b)** 8 **(c)** 10 **(d)** 7

**8.** The perimeter of a rectangle of length $20$ cm and breadth $11$ cm is (cm): 1

**(a)** 61 **(b)** 64 **(c)** 62 **(d)** 63

**9.** A fair coin is tossed once. The probability of getting a tail is: 1

**(a)** $\frac{2}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{4}$

**10.** The common difference of the AP $10, 15, 20, 25, \ldots$ is: 1

**(a)** $4$ **(b)** $7$ **(c)** $5$ **(d)** $6$

**11.** The angle subtended by an arc at the centre is $64^\circ$. The angle it subtends at a point on the remaining part of the circle is: 1

**(a)** $52^\circ$ **(b)** $42^\circ$ **(c)** $32^\circ$ **(d)** $22^\circ$

**12.** The perimeter of a square of side $13$ cm is (cm): 1

**(a)** 54 **(b)** 53 **(c)** 52 **(d)** 51

**13.** A bag has 4 red, 6 blue and 5 green balls. The probability of drawing a green ball is: 1

**(a)** $\frac{1}{6}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

**14.** The next term of the AP $9, 13, 17, 21, \ldots$ is: 1

**(a)** $25$ **(b)** $27$ **(c)** $26$ **(d)** $24$

**15.** An arc subtends $35^\circ$ at a point on the circle. At the centre it subtends: 1

**(a)** $60^\circ$ **(b)** $90^\circ$ **(c)** $70^\circ$ **(d)** $80^\circ$

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** A coin is tossed 60 times and shows heads 33 times. Find the experimental probability of heads and of tails. 2

**17.** Find the common ratio of the GP $2, 10, 50, \ldots$ and its fourth term. 2

**18.** An arc of a circle subtends $84^\circ$ at the centre. Find the angle it subtends at a point on the remaining part of the circle. 2

**19.** Find the circumference of a circle of radius $42$ cm (take $\pi = \frac{22}{7}$). 2

**20.** A fair die is rolled. Find the probability of getting a factor of 6. 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** A bag contains 5 red and 4 blue balls. A ball is drawn at random. Find $P(\text{red})$ and $P(\text{not red})$. 3

**22.** Find the 10th term of the AP with first term $5$ and common difference $2$. 3

**23.** Points $A$ and $B$ lie on a circle with centre $O$ and $\angle AOB = 66^\circ$. A point $C$ lies on the major arc. Find $\angle ACB$ and say why $\angle ADB = \angle ACB$ for another point $D$ on the major arc. 3

**24.** Find the area of a triangle with sides $5$ cm, $12$ cm and $13$ cm using Heron's formula. 3

OR

A die was thrown 60 times and the number 6 appeared 8 times. Find the experimental probability of getting 6, and of not getting 6. 3

**25.** Find the sum of the first 30 natural numbers. 3

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 100^\circ$ and $\angle B = 75^\circ$. Find $\angle C$ and $\angle D$. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** A chord of length $18$ cm is at a distance of $12$ cm from the centre of a circle. Find the radius. 5

OR

A rectangular field is $60$ m long and $35$ m wide. Find its perimeter and area, and the cost of fencing it at ₹15 per metre. 5

**27.** A bag contains 4 red, 5 blue and 6 green balls. Find $P(\text{red})$, $P(\text{blue})$, $P(\text{green})$ and show that they add up to $1$. 5

OR

A sequence is given by $t_1 = 7$ and $t_{n+1} = t_n + 5$. Write the first five terms, say what kind of sequence it is and find $t_{10}$. 5

**END OF QUESTION PAPER** **Total: 50 Marks**

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